The Geopolitics of Compute and the AI Energy Crisis

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Let’s do an Icebreaker!

What do we all actually know about the physical cost of AI?



Let’s discuss the Invisible Hand.



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While software wants to be boundless, AI is constrained by physical hardware.



If software claims to be borderless, what happens when the hardware running it depends entirely on sovereign states?



Let’s focus on the Computer Bottleneck.



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To maintain a strategic edge, the US has restricted the sale of manufacturing equipment to China, and subsidizing domestic fabs (i.e. TSMC to Arizona)



If 90% of advanced AI chips depend on a single island, is AI leadership a computer science problem or a political one?



Now, let’s talk about the Energy Crisis.



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This is already hitting consumers: wholesale electricity costs have spiked up to 267% near dense data center hubs.



When AI infrastructure drives up public utility rates, should tech companies be forced to build their own dedicated power grids?



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Amazon is funding the feasibility and development of SMRs for AWS.



So in short…

Startups are exploring hydrogen-powered data centers…



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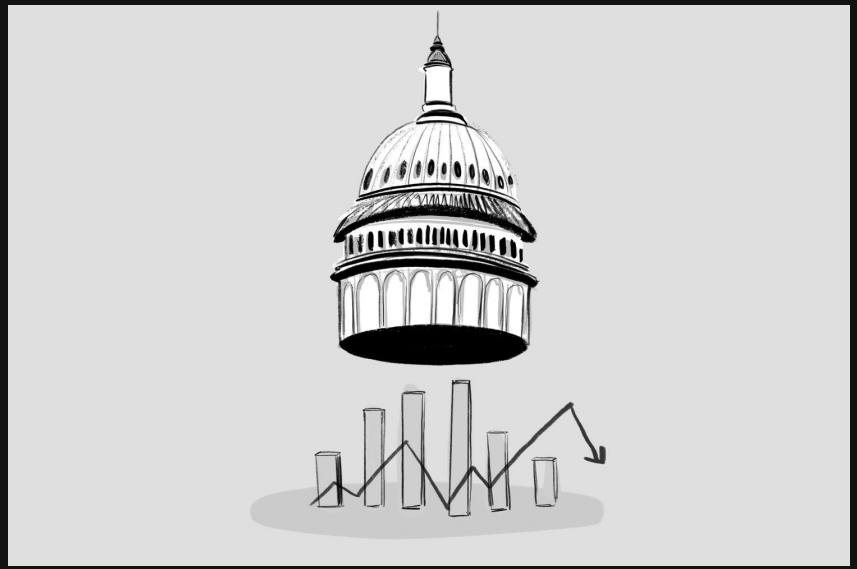
Startups are exploring hydrogen-powered data centers in order to bypass the traditional power grid entirely to deploy facilities twice as fast.



Let’s ask another question…

Should governments heavily regulate where and how AI data centers are built to protect national energy grids?





A useful podcast by MSU CMSE faculty:

Podcast: What are we even doing? (by Dr. Devin Silvia and Dr. Danny Caballero)
Episode: 3 – Why We Can't Have Nice Things (With Prof. Brian O'Shea)



Upcoming MLH event:

SpartaHack is collaborating with MLH to plan the East Lansing Hacktoberfest Meetup.

You can earn Spartacoins to use at SpartaHack 12 in February by attending this event, or by unlocking the achievements on the SpartaHack and the HacktoberfestEL website. Go to spartahack.com/hacktoberfest to sign up!





Sources:

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